

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Data Sheets for Barred Tee
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008
Document Revision	R01
Document Status	Issued For Review
Originator	S.Ikechukwu
Security Classification	Restricted
Issue Date	7 Jun 2022

Data Sheets for Barred Tee

R01	07.06.22	Issued For Review	S.Ikechukwu	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

R01	07.06.22	Issued for Review	S. Ikechukwu	E. Nwachukwu			
Rev No	Date	Reason for Issue / Change	Prepared	Reviewed	TCE Approval	BFPCL Approval	Approver (BFPCL)

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES	
Document No.	Description

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5. Data Sheet for Barred Tees - Fabrication Details

Data Sheet for 24" x 24" Barred Tees- BBOU

Project Name	FEED for Gas Gathering Pipelines & Associated Infrastructure-Phase 1		
Document Title	Data Sheet for Barred Tees	Revision Code.	R01
Document No.	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008	Revision Status	Issued For Review
General Information			
Pipeline	Unit	Parameters	
Pipeline Description		24" Pipeline from OPUG to BBOU. 24" Pipeline from BBOU to ELEP.	
Location		BBOU Manifold Platform	
Design life	Years	30	
Process fluid		Hydrocarbon Gas and Condensate	
Service		Corrosive -CO2	
Type of Pipeline		Swamp / Onshore	
Pipeline Wall Thickness	mm	15.88	
Material Grade		API 5L X70 PSL 2	
Reducing Barred Tee Material		ASTM A860 WPHY 70	
Dimensions			
Run Pipe Normal Pipe Size	In.	24"	
Run Pipe Outside Diameter	mm	610	
Run Pipe Wall Thickness	mm	15.88	
Branch Pipe Normal Pipe Size	In.	24"	
Branch Pipe Outside Diameter	mm	610	
Branch Pipe Wall Thickness	mm	26.2	
Quantity		2 Nos.	
Type		Equal Barred Tee	
Requirements			
Installation		Above Ground Installation	
Connection to Pipe		Girth welding	
End Connection		Bevelled	
Design Conditions			
Design Pressure	MPa/barg	11/110	
Design Temperature	°C	60	
Design factor		0.6	
Corosion Allowance	mm	3	
Pressure Rating		900#	
Fabrication method		Seamless	
SMYS	Mpa	485	
Reference Specification			
Ref.	Document Number	Title	
1	DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems	
2	MSS SP-75	Specification for High Test, Wrought, Butt-Welding Fittings	
3	ASME B31.8	Gas Transmission and Distribution Piping Systems	
4	ASME B16.9	Factory-Made Wrought Buttwelding Fittings	
5	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008	Specification for Barred Tees	
Notes:			
1.	Run Pipe Design Code: ASME B31.8		
2.	Branch Pipe connection is 24" API 5L Gr X60 (WT 26.20mm) Design Code: ASME B31.3		
3.	Tee end with buttwelding type according to ASME B16.25		

Data Sheet for 24" x 24" Barred Tees- ELEP

Project Name				FEED for Gas Gathering Pipelines & Associated Infrastructure-Phase 1			
Document Title				Data Sheet for Barred Tees			
Document No.				EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008			
Revision Code.				R01			
Revision Status				Issued For Review			
General Information							
Pipeline		Unit		Parameters			
Pipeline Description				24" Pipeline from BBOU to ELEP.			
Location				ELEP Manifold Platform			
Design life		Years		30			
Process fluid				Hydrocarbon Gas and Condensate			
Service				Corrosive -C02			
Type of Pipeline				Swamp / Onshore			
Pipeline Wall Thickness		mm		15.88			
Material Grade				API 5L X70 PSL 2			
Reducing Barred Tee Material				ASTM A860 WPHY 70			
Dimensions							
Run Pipe Normal Pipe Size		In.		24"			
Run Pipe Outside Diameter		mm		610			
Run Pipe Wall Thickness		mm		15.88			
Branch Pipe Normal Pipe Size		In.		24"			
Branch Pipe Outside Diameter		mm		610			
Branch Pipe Wall Thickness		mm		26.2			
Quantity				1Pc.			
Type				Equal Barred Tee			
Requirements							
Installation				Above Ground Installation			
Connection to Pipe				Girth welding			
End Connection				Bevelled			
Design Conditions							
Design Pressure		MPa/barg		11 / 110			
Design Temperature		°C		60			
Design factor				0.6			
Corosion Allowance		mm		3			
Pressure Rating				900#			
Fabrication method				Seamless			
SMYS		Mpa		485			
Reference Specification							
Ref.	Document Number			Title			
1	DEP 31.40.10.13-Gen			Design of Pipeline Pig Trap Systems			
2	MSS SP-75			Specification for High Test, Wrought, Butt-Welding Fittings			
3	ASME B31.8			Gas Transmission and Distribution Piping Systems			
4	ASME B16.9			Factory-Made Wrought Buttwelding Fittings			
5	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008			Specification for Barred Tees			
Notes:							
1. Run Pipe Design Code: ASME B31.8 2. Branch Pipe connection is 24" API 5L Gr X60 (WT 26.20mm) Design Code: ASME B31.3 3. Tee end with buttwelding type according to ASME B16.25							

Data Sheet for 36" X 36" Barred Tees- ELEP

Project Name FEED for Gas Gathering Pipelines & Associated Infrastructure-Phase 1			
Document Title Data Sheet for Barred Tees		Revision Code.	R01
Document No. EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008		Revision Status	Issued For Review
General Information			
Pipeline	Unit	Parameters	
Pipeline Description		36" Pipeline from NUN to BBOU. 36" Pipeline from ELEP to AKAB.	
Location		ELEPA Manifold Platform	
Design life	Years	30	
Process fluid		Hydrocarbon Gas and Condensate	
Service		Corrosive -C02	
Type of Pipeline		Swamp / Onshore	
Pipeline Wall Thickness	mm	23.83	
Material Grade		API 5L X70 PSL 2	
Reducing Barred Tee Material		ASTM A860 WPHY 70	
Dimensions			
Run Pipe Normal Pipe Size	In.	36"	
Run Pipe Outside Diameter	mm	914	
Run Pipe Wall Thickness	mm	23.83	
Branch Pipe Normal Pipe Size	In.	36"	
Branch Pipe Outside Diameter	mm	914	
Branch Pipe Wall Thickness	mm	38.2	
Quantity		2 Nos.	
Type		Equal Barred Tee	
Requirements			
Installation		Above Ground Installation	
Connection to Pipe		Girth welding	
End Connection		Bevelled	
Design Conditions			
Design Pressure	MPa/barg	11/110	
Design Temperature	°C	60	
Design factor		0.6	
Corrosion Allowance	mm	6	
Pressure Rating		900#	
Fabrication method		Seamless	
SMYS	Mpa	485	
Reference Specification			
Ref.	Document Number	Title	
1	DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems	
2	MSS SP-75	Specification for High Test, Wrought, Butt-Welding Fittings	
3	ASME B31.8	Gas Transmission and Distribution Piping Systems	
4	ASME B16.9	Factory-Made Wrought Buttwelding Fittings	
5	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008	Specification for Barred Tees	
Notes:			
- Run Pipe Design Code: ASME B31.8 - Branch Pipe connection is 36" API 5L Gr X60 (WT 38.20mm) Design Code: ASME B31.3 - Tee end with buttwelding type according to ASME B16.25			

Data Sheet for 36" X 36" Barred Tees- AKABEL

Project Name	FEED for Gas Gathering Pipelines & Associated Infrastructure-Phase 1		
Document Title	Data Sheet for Barred Tees	Revision Code.	R01
Document No.	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008	Revision Status	Issued For Review
General Information			
Pipeline	Unit	Parameters	
Pipeline Description		36" Pipeline from ELEP to AKAB.	
Location		AKAB. Station	
Design life	Years	30	
Process fluid		Hydrocarbon Gas and Condensate	
Service		Corrosive -CO2	
Type of Pipeline		Swamp/Onshore	
Pipeline Wall Thickness	mm	23.83	
Material Grade		API 5L X70 PSL 2	
Reducing Barred Tee Material		ASTM A860 WPHY 70	
Dimensions			
Run Pipe Normal Pipe Size	In.	36"	
Run Pipe Outside Diameter	mm	914	
Run Pipe Wall Thickness	mm	23.83	
Branch Pipe Normal Pipe Size	In.	36"	
Branch Pipe Outside Diameter	mm	914	
Branch Pipe Wall Thickness	mm	38.2	
Quantity		1 No.	
Type		Equal Barred Tee	
Requirements			
Installation		Above Ground Installation	
Connection to Pipe		Girth welding	
End Connection		Bevelled	
Design Conditions			
Design Pressure	MPa/barg	11/110	
Design Temperature	°C	60	
Design factor		0.6	
Corosion Allowance	mm	6	
Pressure Rating		900#	
Fabrication method		Seamless	
SMYS	Mpa	485	
Reference Specification			
Ref.	Document Number	Title	
1	DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems	
2	MSS SP-75	Specification for High Test, Wrought, Butt-Welding Fittings	
3	ASME B31.8	Gas Transmission and Distribution Piping Systems	
4	ASME B16.9	Factory-Made Wrought Buttwelding Fittings	
5	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008	Specification for Barred Tees	
Notes:			
1.	Run Pipe Design Code: ASME B31.8		
2.	Branch Pipe connection is 36" API 5L Gr X60 (WT 38.20mm) Design Code: ASME B31.3		
3.	Tee end with buttwelding type according to ASME B16.25		

Data Sheet for Barred Tees - Fabrication Details

